

130 THE CONSTRUCTION OF THE SUEZ CANAL

weeks. But by 1866, a new cable had been installed, capable of transmitting 120 words per minute. By 1870, a telegraph link had been established between London and Mumbai; this was then extended to Australia in 1872 and New Zealand in 1876. By 1902, the United States was linked to Hawaii. This was the first near-instant international communications system.

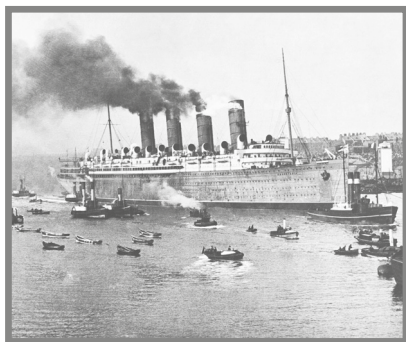
The Great Eastern

The ship responsible for laying the transatlantic cable in 1866 was the *Great Eastern*, designed by the most visionary engineer of the first phase of the Industrial Revolution, Isambard Kingdom Brunel. Designed to carry 4,000 passengers from England to Australia nonstop (and to return to England without refueling), the ship was overly ambitious in concept and a commercial failure.

However, it was indicative of a trend toward larger, faster, and safer ships. Unlike the *Great Eastern*, which was built of iron, later, steel-built, propeller-driven ships would prove more versatile. Their introduction coincided with the development of more powerful and efficient steam engines.

Steamships and trade

The decline of the sailing ship further transformed imperial trade. One notable



The RMS Mauretania, built at Wallsend, Tyne and Wear, UK, was the largest and fastest ship in the world. In 1909, it set a record, sailing across the Atlantic in less than five days.

result was the introduction of a series of ever-larger passenger ships. The transatlantic route saw the most obvious developments. In 1874, the British steamer *Britannic*, capable of generating 5,500 horsepower, set a new east–west Atlantic record of just under eight days. In 1909, the *Mauretania*, which generated 70,000 horsepower and carried over 2,000 passengers, set a new record of four days and 10 hours, cruising at an average speed of 26 knots, or 30 mph (48 km/h).

The Great Mineral Rush

The search for new sources of minerals, both precious and industrial, reached new heights toward the close of the 19th century. Discoveries of diamonds and gold in the US, Canada, Australia, and—most significantly of all—South Africa sparked a frenzy of development. Diamonds were discovered in South Africa's Orange Free State in 1867, and gold in the Transvaal in 1886. Both were independent

Boer republics, established by the descendants of the original Dutch settlers of what had become the British Cape Colony. Their heightened economic importance reinforced Britain's determination to annex them, which they could do only after the bitter Boer War (1899–1902), which stretched Britain's military resources to their limits. The exploitation, both before and after the conflict, of the mineral resources of what in 1910 became the Union of South Africa by armies of underpaid black workers would later prove

to be critical in the institutionalizing of Apartheid.



Working conditions in South Africa's gold mines were harsh, and the work force—mainly young black men—was exploited and underpaid.